

Bonus Task 3: Alternative Motors with Better thrust-per-watt

T-Motor Air Gear 2216 (KV900) – same footprint/mounting as the Holybro motor (literally the same motor family), but T – Motor publishes full thrust/current/efficiency tables, letting us cite grams of thrust per watt directly instead of estimating. This is a genuine drop in swap.

SunnySky X2216 V3 (KV880-900) -- published bench data shows ~12+ g/W efficiency at low-to-mid throttle (i.e. near hover), dropping off at full throttle — which is exactly the regime that matters for endurance, since the drone spends nearly all its flight time near hover, not at max throttle.

The efficiency argument is that a motor's grams per watt figure is highest at partial throttle and falls as you approach max thrust. Since hover and not max throttle is what determines endurance, the right optimization target is hover-throttle efficiency, not peak thrust. A motor advertised for "better thrust" that's actually less efficient at the ~35–40% throttle this build hovers at would hurt endurance, not help it.